

Labor market reform as an external enabler of high-growth entrepreneurship: A multi-level institutional contingency perspective

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ABSTRACT

We investigate the impact of friction-reducing labor market reforms on regional high-growth entrepreneurship (HGE) through the effects of reduced legal enforceability of noncompete agreements (NCAs). We draw on new institutional economic theory and the external enablement framework, with insights from the theory of market-preserving federalism, to explore how these reforms enable (disable) HGE within the context of other, concurrent institutional changes at different governance levels. We assemble a novel multi-level longitudinal dataset and employ staggered difference-in-differences estimation to assess causal effects. Our findings suggest that while reducing the enforceability of NCAs can foster regional HGE, the effectiveness of such reforms is heavily influenced by concurrent federal and local institutional changes. In sectors facing significant federal regulatory expansion, the benefits brought by the reduction of NCA enforceability are negated. However, local pro-market institutional changes can counteract the disabling effects of federal regulatory expansion. This highlights the need to consider how the evolving institutional environment influences potential enablers of HGE, cautioning against claims that these labor market reforms (or other exogenous environmental changes) universally yield positive entrepreneurship outcomes.

Executive summary

Entrepreneurship scholars have paid scarce attention to the role of labor market institution. Although existing research highlights that non-compete agreements (NCAs), employment contracts that restrict employees from working for or starting a competing business, such studies have predominantly focused on their impact on new venture creation rather than their influence on the subsequent growth of entrepreneurial ventures. We address this gap by investigating how state-level reforms that weaken NCA enforceability affect HGE and how the impact of these labor market reforms interacts with federal and local institutional conditions.

Using a multi-level institutional framework, we integrate insights from new institutional economic theory, market-preserving federalism, and the external enablement (EE) framework to develop a comprehensive model of how federal regulatory expansion and local pro-market institutional changes moderate the effects of NCA reforms on HGE. Employing staggered difference-in-differences

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estimation across 15 industries and 270 metropolitan statistical areas in the U.S., we reveal that weakening NCAs leads to a meaningful increase in HGE. However, this positive effect is weakened by federal regulatory expansion, which introduces substantial compliance burdens that negate the benefits of labor market reforms. In contrast, local pro-market institutional changes offset the negative impact of federal regulations.

Our study contributes to theory by extending the EE framework to high-growth entrepreneurship, introducing novel disablement mechanisms (resource escalation and elongation), and illustrating the importance of multi-level institutional interactions in shaping entrepreneurial outcomes. For practitioners, these findings underscore the need to understand the broader institutional landscape when assessing potential growth strategies. For policymakers, our findings demonstrate that local pro-market reforms can effectively counterbalance negative effects imposed by federal regulatory expansion, providing local policy makers a bottom-up policy approach to foster high-growth entrepreneurship despite top-down regulatory constraints imposed by federal policymakers.

1. Introduction

Entrepreneurship scholars widely acknowledge that institutions matter (Bjørnskov and Foss, 2016; Boudreaux et al., 2019; McMullen et al., 2008), but surprisingly they “have expressed little interest” (Burton et al., 2019, p. 1054) in examining labor market institutions, which govern the functioning of the labor market and shape individual incentives to become an entrepreneur or work for a startup. One exception is the nascent literature examining how non-compete agreements (NCAs), contractual clauses that restrict employees from working with competing business or starting their own ventures in the same field for a certain period after leaving their job (Marx et al., 2009), create frictions in the labor market that constrain employee mobility and reduce entrepreneurial entry (Campbell et al., 2017). Because NCAs are quite common in practice—a recent survey of the U.S. labor force found that 18 % are currently bound to an NCA and 38 % were subject to one in the past (Starr et al., 2021)—their potential effects on entrepreneurship are significant. Additionally, NCAs are particularly common among high-skilled employees (Garmaise, 2011; Jeffers, 2024; Kini et al., 2020; Marx, 2011; Starr et al., 2021), who may possess important industry knowledge and strategic human capital that is key for successful entrepreneurship (Dwyer and Kotey, 2016). Taking advantage of the variance in the enforceability of NCAs across geographic spaces (e.g., different states within the U.S.), scholars have identified a link between the legal enforceability of NCAs and reduced levels of entrepreneurial entry (Buenstorf et al., 2016; Jeffers, 2024; Marx, 2021; Samila and Sorenson, 2011), suggesting that NCAs can increase the costs and risks faced by employees considering starting or joining a new business (Rauch, 2016; Starr et al., 2018).

Despite the growing number of studies examining the link between reduced legal enforceability of NCAs and entrepreneurial entry, there has been an almost exclusive focus on the *triggering* role (i.e., how it affects new venture creation; Davidsson et al., 2020) of this institutional change. In contrast, the ability of such institutional reforms to affect other stages of entrepreneurial activity, such as the shaping of a venture, its processes, offerings, and, ultimately, its growth and scaling, has been overlooked. To the best of our knowledge, no studies have examined how diminishing the legal enforceability of NCAs influences *high-growth* entrepreneurship (HGE) in a region. This consideration is relevant for several reasons. First, HGE is a regional phenomenon and there is a need to better understand how geographic factors, including institutional changes, influence the regional heterogeneity of HGE (Araki et al., 2024; Sleuwaegen and Ramboer, 2020). Accordingly, a reduction in the enforceability of NCAs can represent an important type of friction-reducing labor market reform because it can facilitate the flow of talent, knowledge, and specific information, potentially compressing the time and conserving the resources otherwise necessary for these flows to occur. Thus, the possibility that NCAs may not only hamper new business formation but also limit the ability of entrepreneurial ventures to strategically hire high-skill employees to facilitate growth (Campbell et al., 2017; Ewens and Marx, 2018) motivates our research question: *How do friction-reducing labor market reforms, specifically the diminished legal enforceability of NCAs, influence local high-growth entrepreneurship?*

To address this question, we build on the catalog of concepts advanced in the external enabler (EE) framework (Davidsson et al., 2020) and combine it with new institutional economic theory (NIE) (North, 1990) and insights from market-preserving federalism (MPF) theory (Weingast, 1995) to develop a multi-level institutional contingency model. We first contend that the diminished enforceability of NCAs is a friction-reducing labor market reform that enables local high-growth entrepreneurship. Second, based on NIE, we suggest that the HGE-enabling effects of reduced NCA enforceability are concentrated in industries facing lower levels of regulatory expansion. Finally, drawing inspiration from MPF theory, we suggest that the disabling effects in industries facing a growing regulatory burden may be offset by local pro-market institutional changes (Lucas and Boudreaux, 2020).

To test our conceptual model, we assembled a novel multi-level longitudinal dataset comprising 15 different industries, nested within 270 U.S. metropolitan statistical areas (MSAs), spanning the period 2004 to 2017. Exploiting a natural experiment involving state-level weakening of NCA enforceability, we employ staggered difference-in-differences (DID) estimation (Cengiz et al., 2019; Goodman-Bacon, 2021) to estimate the causal effect of these state-level labor market reforms on local HGE. We further examine the heterogeneity of these effects across industries by testing for multi-level institutional contingencies. Consistent with our hypotheses, we find robust evidence that diminished NCA enforceability causes an economically meaningful increase in HGE, but the positive effects of friction-reducing NCA reform are weakened in industries facing greater federal regulatory expansion. Furthermore, the negative effect of growing federal regulation is offset by local pro-market institutional changes.

Our study makes important contributions to several streams of literature. First, we contribute to the NCA-entrepreneurship literature by showing that reduced NCA enforceability not only facilitates venture creation but also influences venture *growth*, contingent on institutional conditions across various levels of governance. More generally, we contribute to the institutions and entrepreneurship literature by providing further support for the emerging institutional contingency perspective by illustrating how multi-level institutional changes shape the geography of high-growth entrepreneurship. Specifically, our study advances the nascent

MPF approach in entrepreneurship by extending the focus from two interdependent institutional levels, i.e., federal regulation and subnational economic policy (Lucas and Boudreaux, 2020), to account for a third level of institutional interdependence, i.e., judiciary reform. We also extend the EE literature, which has primarily focused on new venture creation, towards high-growth entrepreneurship, and offer novel insights into how concurrent environmental changes interact across levels to not only enable but also disable entrepreneurship.

2. Theoretical framework

In this section, we describe the conceptual and theoretical foundations in our theorizing that links multi-level institutional changes with regional high-growth entrepreneurship (HGE). Starting with the vernacular provided by external enabler (EE) framework (Davidsson et al., 2020), we incorporate principles from new institutional economics theory (North, 1990), and combine them with insights from market-preserving federalism (MPF) theory (Weingast, 1995) to develop a multi-level institutional contingency model.

Our model, depicted in Fig. 1, illustrates our three-layered conceptual framework. First, we hypothesize that labor market reforms reducing NCA enforceability act as an external enabler, leading to an increase in HGE (H1). However, we posit that this main effect is weakened by federal regulatory expansion, which we associate with disabling mechanisms—*elongation and resource escalation*—of entrepreneurship (H2). Finally, drawing inspiration from the MPF theory, we suggest that the negative impacts of federal regulatory expansion are mitigated by local pro-market institutional changes (H3).

In the next sections, we define our key constructs, including HGE and NCAs, briefly describe the EE framework, and provide an overview of NIE and MPF in the context of entrepreneurship to set the stage for our theory development.

2.1. High-growth entrepreneurship (HGE)

HGE is defined as the process of rapidly developing an entrepreneurial venture (Araki et al., 2024; Dwyer and Kotey, 2016; Kiviluoto, 2013; Sleuwaegen and Ramboer, 2020). Thus, conceptually, HGE focuses on the ability of a business to scale up its operations significantly within a short period in a way that is well above average industry benchmarks (Delmar et al., 2003). In the emerging geography of high-growth entrepreneurship literature, HGE represents a subset of entrepreneurial activity in a region characterized by rapid and significant growth in key business metrics, notably employment or revenue (Araki et al., 2024; Fotopoulos, 2022; Sleuwaegen and Ramboer, 2020; Venâncio et al., 2020). Note that this conceptualization of HGE is not dependent on the age of a business and is sufficiently general to capture both startups and existing businesses that experience rapid growth.

2.2. Non-compete agreements (NCAs)

NCAs are common contractual clauses that restrict employees from working with competing businesses or starting their own ventures in the same field for a certain period after leaving their job (Samila and Sorenson, 2011; Starr, 2019). These clauses are particularly prevalent among high-skill, high-pay workers, who possess strategic human capital valuable to competitors or for initiating new ventures (Jeffers, 2024; Marx et al., 2009; Starr et al., 2021). The primary purpose of NCAs is to protect a firm's investments in human capital, intellectual property, and interfirm relations by limiting employees' ability to compete with their current employer (Conti, 2014; Garmaise, 2011; Starr, 2019). However, they also introduce frictions that hinder within-industry labor mobility (Lipsitz and Starr, 2022; Ransom, 2021). For instance, employees bound by NCAs expect that they will face significant financial costs (e.g., costs to buyout the contract or resolve legal disputes) to violate these contracts and join a competing business or create a new venture in the same industry (Starr et al., 2020). NCAs may also create externalities by affecting the mobility of workers not directly covered by them (Starr et al., 2019).

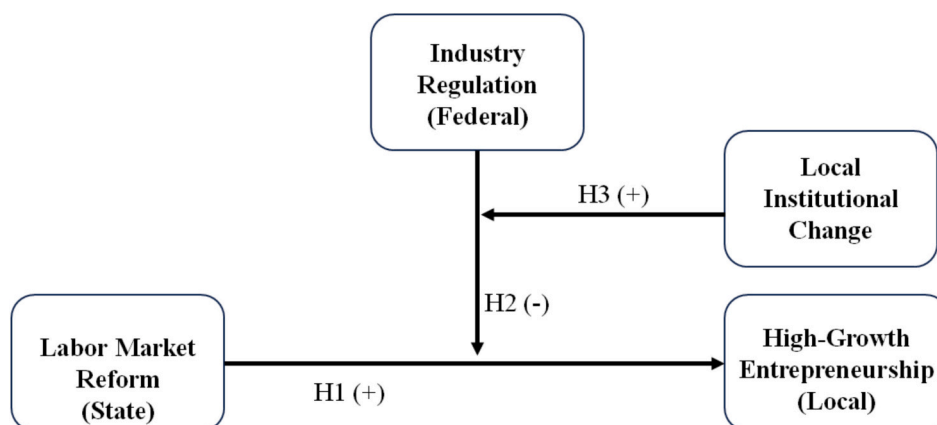


Fig. 1. Conceptual model.

2.3. The external enablement (EE) framework

The EE framework focuses on disequilibrating environmental changes—such as technological change (Schade and Schuhmacher, 2022; von Briel et al., 2018), infrastructure development (Bennett, 2019; Chen et al., 2020), and institutional change (Bennett, 2021a; Lucas et al., 2023)—that create entrepreneurial opportunities for some actors. The EE framework conceptualizes such changes in terms of their characteristics, mechanisms, and roles. Characteristics such as the scope (sectoral, spatial, temporal, or socio-demographic) and onset (suddenness and predictability) define the nature and impact of an EE. Mechanisms are the cause-effect relational constructs linking environmental changes to entrepreneurial outcomes by influencing key factors such as cost, time, and functionality, even if they are not directly observable. The roles of an EE—triggering, outcome-enhancing, and shaping—characterize specific functions or effects that enablers have within the entrepreneurial process. Although existing EE studies primarily focus on new venture creation—characterized by the triggering role—the EE framework also highlights the outcome-enhancing and shaping roles as important elements to understand the performance and the strategic effects of EEs on existing ventures (Davidsson et al., 2023; Kimjeon and Davidsson, 2021).

2.4. The role of institutions at different governance levels

To theorize how friction-reducing labor market reforms such as diminished enforceability of NCAs influence HGE, we employ NIE theory, which defines institutions as “humanly devised constraints that structure political, economic, and social interaction,” encompassing both formal institutions (i.e., laws, regulations, and policies) and informal institutions (i.e., cultural norms, values, and beliefs) (North, 1991, p. 97; Williamson, 2000). Our focus is predominantly on formal institutions and their impact on the entrepreneurial landscape.¹ Institutions, particularly formal ones, play a critical role in shaping the “rules of the game” for entrepreneurs by influencing the transactions costs underpinning exchange (Baumol, 1990). This encompasses elements such as labor laws, taxation policies, property rights, and industry-specific regulations (Bailey and Thomas, 2017; Sobel, 2008). Indeed, a growing body of literature corroborates that formal institutions, by shaping transactions costs, matter for entrepreneurship (Bjørnskov and Foss, 2016). Studies have shown, for instance, that variations in the formal institutional environment at the country level impact entrepreneurial entry (Boudreaux et al., 2019; Gohmann, 2012; McMullen et al., 2008), with specific country-level institutional changes acting as catalysts for HGE (Eberhart et al., 2017; Eesley, 2016). Moreover, scholars recognize that entrepreneurship is a largely regional phenomenon and that there are important regional institutional differences that influence entrepreneurial activity (Bennett, 2021a, 2021b; Coomes et al., 2013), including regional HGE (Araki et al., 2024; Venâncio et al., 2020).

Institutions operate at different levels of governance and, importantly, they may not always be in alignment with each other (Williamson, 2000). In our study, we consider three levels of institutional governance—federal, state, and local—and analyze their relationship. MPF theory provides a valuable perspective for examining how decentralized governance systems interact to influence local economic activity. Central to MPF theory is autonomy at lower levels of government to enact institutional reforms to govern local economic activity so long as they do not directly contradict the rules established at higher levels of government. There are three essential conditions for effective subnational (i.e., state and local) governance: (1) the predominance of subnational institutions in regulating local economic activities, (2) the assurance of unrestricted inter-jurisdictional trade forming a common market, and (3) the imposition of hard budget constraints on subnational governments (Qian and Weingast, 1997; Weingast, 1995). The institutional autonomy ascribed to policymakers at the subnational level, who “possess local knowledge and accountability to their particular economic context” (Lucas and Boudreaux, 2020, p. 1), provides them with the ability to enact institutional reforms that counteract or enhance the economic effects of institutional reforms emanating from higher levels of governance. As such, MPF theory provides a helpful lens to investigate the interdependence of institutional reforms at various governance levels and how it can generate diverse economic outcomes by influencing the ability of individuals to benefit from potential enablers of entrepreneurship.

2.5. Reduced NCA enforceability & high-growth entrepreneurship

Our first hypothesis suggests that a labor market reform that diminishes the legal enforceability of NCAs can stimulate regional high-growth entrepreneurship. Our theorizing is rooted in understanding that high labor mobility is indicative of a dynamic and flexible labor market where talent and resources can be more efficiently reallocated. In this context, NCAs may generate constraints to labor market mobility by reducing the ease with which employees can change jobs and hampering the ability of the workforce to respond to changing market demands, pursue new opportunities, or bring their skills and experiences to different organizational contexts (Campbell et al., 2017). When institutionalized, NCAs limit the ability of employees to compete with their current employers,

¹ Some readers may be more familiar with institutional theory (IT) than NIE. The former is more common in sociology and organizational theory, while the latter is more common in economics and political science. While both theories emphasize how individual and organizational behavior is shaped by the institutional environment, their underlying assumptions differ. While NIE focuses more on economic rationality and utility maximization within institutional constraints, it places a stronger emphasis on the pursuit of legitimacy and conformity to social norms and expectations. Another difference between the two frameworks is that IT scholars typically employ Scott's (1995) three-pillar institutional framework (i.e., regulative, normative, and cognitive), while NIE scholars adopt the formal-informal institutional nexus (North, 1991; Williamson, 2000). However, in examining the micro-foundation of cognitive institutions, Boudreaux et al. (2023) lay the groundwork for a potential reconciliation of the two institutional perspectives. Such reforms enhance the availability of skilled workers for entrepreneurial ventures.

creating a labor market that is less responsive to the needs of emerging businesses and entrepreneurial ventures (Ewens and Marx, 2018; Marx et al., 2009). This can result in a less dynamic and competitive economy and in the underutilization of human capital if employees are bound to roles that do not fully leverage their skills or aspirations. Thus, NCAs may act like other types of labor market institutions (e.g., nondisclosure and confidentiality agreements, intellectual property enforcement, occupational licensing regulations, hiring and firing regulations) that also impede labor mobility (Burton et al., 2019; Feldmann, 2009).

In the U.S., the context of our study, the legal enforceability of NCAs has historically been governed by state law, with some states such as California, North Dakota, and Oklahoma prohibiting them and others enforcing them with various degrees of strictness. The legal enforceability of NCAs has been subject to change over time and this has occurred to numerous states in recent decades. For example, within the time frame of our study (between 2004 and 2017), four states—Oregon, Montana, South Carolina, and New Hampshire—weakened the legal enforceability of NCAs, thereby serving as an exogenous state-level institutional change influencing labor mobility (Ewens and Marx, 2018; Jeffers, 2024).² The conceptual framework of external enablers (Davidsson et al., 2020) is well-suited to describe and theorize on the characteristics and potential mechanisms of these institutional changes for regional HGE. Specifically, we suggest that such labor market reforms cause an increase in HGE through three enabling mechanisms: compression, conservation, and human resource expansion.

The first two mechanisms are compression, which entails a reduction in the time required for an activity, and conservation, which refers to a decrease in the resources needed for an activity (Davidsson et al., 2020). In the context of mobility-enhancing labor market reforms such as reduced NCA enforceability, compression can mean reducing the time it takes an employee to transition from an employee to entrepreneur, and conservation can mean lowering switching costs from current employment to an entrepreneurial role. The reduction in both the time and the costs for an employee to transition to entrepreneurship can be viewed collectively as transaction costs reductions (Coomes et al., 2013; Ransom, 2021). Thus, within the context of mobility-enhancing labor market reforms, compression and conservation mechanisms have the common element of transaction cost reduction.

When NCAs are enforceable, prospective entrepreneurs may delay or be deterred from leaving secure employment to start new ventures by the loss of income and time while waiting for the noncompete duration to expire. Labor market reforms weakening NCA enforceability may enable quicker transitions into entrepreneurship by reducing the expected financial, bureaucratic and legal costs associated with overcoming these barriers to mobility, allowing entrepreneurs to conserve valuable time and resources that can be used to plan and finance their ventures (Starr et al., 2018). Additionally, because NCAs are particularly prevalent among high-skill, high-pay workers (Jeffers, 2024; Marx, 2011; Marx et al., 2009; Starr et al., 2021), including executives (Garmaise, 2011; Kini et al., 2020), weakening NCA enforceability can lower the transactions costs of innovative employee spinouts with high-growth potential (Buenstorf et al., 2016; Campbell et al., 2012; Starr et al., 2018).

Furthermore, NCAs can increase the costs for existing entrepreneurial ventures, which often face resource limitations, to recruit skilled workers with relevant industry experience to help fuel growth (Lee, 2014). In addition to providing attractive compensation packages, these costs are amplified by potential buyout and legal costs for prospective employees subject to an NCA (Marx, 2021). Because NCAs constrain the ability of entrepreneurial ventures to strategically hire employees to help facilitate their growth (Marx et al., 2015; Mawdsley and Somaya, 2016), weakening NCA enforceability can also act as a human resource expansion mechanism by increasing the firm's accessibility of valuable human resources (Lucas et al., 2023). Specifically, such reforms enhance the availability of skilled workers for entrepreneurial ventures by lowering their hiring costs and broadening the pool of knowledge workers open to pursuing new opportunities in a growth-oriented venture (Starr et al., 2020), which can also be viewed as a transaction cost reduction.

In sum, the mechanisms identified in the EE framework—compression, conservation, and human resource expansion—can help explain how reduced enforceability of NCAs creates a more conducive environment for HGE. This is achieved by lowering transaction costs related to both time and resources needed by employees to transition into entrepreneurial roles, and by expanding the talent pool available for emerging ventures with high growth potential. Thus, we hypothesize that:

H1. Labor market reforms that reduce legal enforceability of non-compete agreements are associated with an increase in local high-growth entrepreneurship.

2.6. The disabling role of federal regulatory expansion

While we propose that reduced enforceability of NCAs creates a more conducive environment for HGE, the benefits of such institutional changes are unlikely to be uniformly distributed across all sectors (Davidsson et al., 2020), particularly given the heterogeneity of federal regulatory regimes and associated compliance costs across industries (Al-Ubaydli and McLaughlin, 2017). Integrating insights from NIE and the EE framework, our second hypothesis proposes that the expansion of federal regulations serves as a conditional external *disabler*, countering the enabling effects (compression, resource conservation, and human resource expansion) of diminished NCA enforceability. The idea of entrepreneurship-disabling institutions has been introduced in previous research (Bennett,

² Lucas (2024) finds that state-level political regime change, operationalized as displacement of the incumbent party as governor, is associated with a short-term reduction in growth-oriented entrepreneurial entry. To rule out the possibility of political regime change exerting a confounding effect, we verified that there were no displacements coinciding with the four states that weakened NCA enforceability in our sample. We also re-estimated the baseline regressions (Table 3) and included variables to capture divided political control, unified party control, and gubernatorial party changes at the state level. There is no consistent evidence that our HGE measures are correlated with these political variables and including them in the regressions had no meaningful effect on our estimated treatment effects. Results omitted for space but available upon request.

2019, 2021a). However, we extend this notion by analyzing the interrelation of institutional changes at different levels of governance.

Our theory is predicated on the idea that while state-level NCA reforms aim to reduce friction in the labor market by enhancing mobility and decreasing transaction costs, concurrent federal regulatory expansion may elongate the entrepreneurial process and escalate the resources needed for venture development and scaling, thus inhibiting the positive impact of the state-level labor market reform on HGE. Thus, we introduce two novel disabling mechanisms—*elongation* and *resource escalation*—that reflect the impact of increased federal regulatory expansion. These mechanisms, we argue, undermine the positive effects associated with labor market reform by introducing significant transaction costs that can overshadow the savings provided by reduced NCA enforceability.

Elongation, conceptualized as the reverse of compression (Davidsson et al., 2020), refers to the process of extending the time required to complete tasks or activities due to the added complexities, increased requirements, or unforeseen obstacles of an environmental change. In the context of regulatory expansion, elongation can involve an increase in the time and processes required for compliance, extending the time expended to learn and navigate new regulations, and meet enhanced compliance standards.

Resource escalation, conceptualized as the reverse of resource conservation (Davidsson et al., 2020), refers to the increased allocation of resources required to meet expanded or intensified conditions as the result of an environmental change. In the context of federal resource expansion, resource escalation can lead businesses to invest more in compliance activities, including hiring legal experts, dedicating staff for compliance management, and investing in new technologies or systems to meet regulatory standards. Such increased resource commitments can strain the limited resources of startups and growing businesses, diverting resources and focus away from core business activities that drive growth such as product development and market expansion (Long et al., 2022; Minniti and Palubinskas, 2023).

Because federal regulatory expansion involves lengthier times and more complex process for entrepreneurs to launch and grow their ventures, the effect of weakened NCA enforcement on HGE may vary significantly by industry, given that entrepreneurs operating in them face different degrees of regulatory expansion. In industries with low levels of federal regulatory expansion, the time compression enabled by reduced NCA enforceability can be fully realized. Entrepreneurs in such industries face fewer regulatory hurdles and can more quickly navigate the process of starting and growing their ventures (Bailey and Thomas, 2017; Chambers et al., 2022). Likewise, high-skilled employees may more quickly implement their decisions to join a high-growth potential venture, ultimately enabling them to take greater advantage of the opportunity offered by reduced NCA enforceability. Conversely, in industries experiencing high levels of federal regulatory expansion, this may not be the case. Especially for HGE, where timing is critical for capitalizing on technological advances or market opportunities (Araki et al., 2024), the ability to quickly seize an opportunity might be dampened. For instance, expanding federal regulations may introduce extra tasks, requirements, and uncertainty into the entrepreneurial process. Because extra efforts are necessary for navigating these regulatory changes, dynamism can be lost and the decision-making process can be elongated, ultimately slowing down the translation of entrepreneurial opportunities into de facto high-growth entrepreneurship (Wood et al., 2016).

Federal regulatory expansion can also negate the positive impact of diminished NCA enforceability on HGE through resource escalation. With growing regulatory demands, high-growth potential businesses may need to divert resources away from strategic growth initiatives towards compliance-related activities. For instance, they will need greater amounts of financial and human capital to meet new compliance standards. As a result of this regulatory-induced shift in resource allocation, entrepreneurial ventures may not be able to tap into the EE mechanisms that should have been facilitated by the labor market reform. Furthermore, the transaction costs savings enabled by weakened NCA enforceability can be dwarfed by the increased transaction costs associated with regulatory compliance. Meanwhile, the enabling effects of these labor market reforms may not be hampered in sectors facing lower levels of regulatory expansion as entrepreneurial ventures can strategically allocate their resources towards growth initiatives rather than tie them up in compliance. Thus, we hypothesize that:

H2. Federal regulatory expansion weakens the positive relationship between reduced NCA enforceability and high-growth entrepreneurship.

2.7. The compensatory role of local pro-market institutional change

We consider local institutional changes as an additional layer with the potential to moderate the effect of federal regulatory expansion on the ability of a state-level labor market reform to induce HGE. This draws inspiration from market-preserving federalism (MPF) theory, which suggests that lower-level institutions (e.g., state or local) can counteract (enhance) the negative (positive) effects of upper-level (e.g., federal) institutions (Qian and Weingast, 1997; Weingast, 1995). We adapt MPF to examine the interrelation of institutional changes at the federal, state, and local levels. Specifically, we contend that local pro-market institutional changes—defined as changes in the local rules, policies, and regulations that encourage market exchange and restrict governmental interference in the local economy (Bennett, 2021a)—may counteract the disabling mechanisms (i.e., elongation and resource escalation) of federal regulatory expansion, offsetting the negative moderating impact this expansion has on the HGE-enabling role of reduced NCA enforceability. Our theorizing thus builds on MPF theory's central insight that subnational governments can support market dynamism and entrepreneurial growth despite potentially disabling higher-level institutional changes (Lucas and Boudreaux, 2020).

As we theorized above, HGE enablement from reduced NCA enforceability may be disabled in industries facing a growing federal regulatory burden, as the increased transaction costs associated with complying with regulatory expansion may dwarf the transaction costs savings associated with NCA reform. We contend that local pro-market institutional changes, through resource conservation, compression, and enclosing mechanisms (Bennett, 2021a; Kimjeon and Davidsson, 2021), may counteract the disabling effects of federal regulatory expansion attributable to resource escalation and elongation. Local policymakers enacting tax and spending reforms

that allow individuals and businesses to retain a greater portion of their income and resources (Bennett, 2021b) can serve as a conservation mechanism, buffering the resource escalation imposed by federal regulatory expansion (Lucas and Boudreaux, 2020). Similarly, local policymakers enacting labor market reforms that lower the costs of recruiting and hiring employees (Feldmann, 2009) can allow entrepreneurs to conserve resources, again acting as a buffer to the escalation of resource expenditure for growing regulatory compliance (Arif, 2023; Lucas and Boudreaux, 2020). Local pro-market institutional changes may also simplify the process of local institutional compliance (Lucas et al., 2023), acting as a compression mechanism that buffers the elongation associated with a growing federal regulatory burden, enabling entrepreneurs to allocate more of their time and effort to launching and growing their business instead of focusing on compliance issues (Baumol, 1990). Thus, local pro-market institutional changes can create a more favorable environment for HGE through an enclosing mechanism that allows high-growth-potential ventures to focus more on capturing the value they create, thereby better positioning them to benefit from the reduced labor market frictions created by weakened NCA enforceability. Thus, we hypothesize that:

H3. Local pro-market institutional changes offset the negative moderating effect of federal regulatory expansion on the relationship between reduced NCA enforceability and high-growth entrepreneurship.

3. Data & methods

3.1. Research context

The U.S. is the context for our empirical analysis. Because entrepreneurship is a regional phenomenon and following recent regional entrepreneurship studies using the U.S. as the context (Bennett, 2021a; Coomes et al., 2013; Wagner and Bologna Pavlik, 2020), we use the metropolitan statistical area (MSA) as our unit of analysis. Our main independent variable—friction-reducing labor market reforms—occurs at a higher level of governance, i.e., the state. We observe, therefore, these institutional changes at the state level. MSAs are nested within states, providing us with a multi-level dataset (Lucas and Boudreaux, 2020). We account for the nested nature of our data by clustering the standard errors at both the state and MSA level (Cameron et al., 2011). Our final sample consists of 270 MSAs over the period 2004–2017. The period of our analysis is limited by data availability. Table 1 provides descriptions and summary statistics for the variables included in our analysis. Table 2 provides a correlation matrix.

3.2. High-growth entrepreneurship (HGE)

While growth is a heterogenous concept that may be based on numerous indicators (Kiviluoto, 2013; Shepherd and Wiklund, 2009), in terms of methodology, two main approaches of gauging HGE have emerged in the literature. The first approach is based on the proportion of businesses in a population that experience the highest growth over a particular period (e.g., the 5 % of fastest-growing businesses). The second defines high-growth in terms of a threshold pace of growth over a specific period (e.g., growth rate of employment or sales over a three-year period) (Coad et al., 2014). We use both approaches in our study. Following recent research, we operationalize our dependent variable—high-growth entrepreneurship—based on the prevalence of high-growth businesses within a geographic territory (Araki et al., 2024; Friesenbichler and Hölzl, 2020; Sleuwaegen and Ramboer, 2020). Because our analysis involves multiple geographic levels, we operationalize HGE at the lowest level (i.e., local or MSA-level). Although there are a multitude of high-growth measures that capture different dimensions of growth (Delmar et al., 2003), the two most common are based on employee and revenue growth (Kiviluoto, 2013; Shepherd and Wiklund, 2009). Due to data limitations, our primary measure is based on employment growth.

Our primary measure is derived from the National Establishment Time Series (NETS) database, which is a rich source of establishment-level data with national coverage. NETS was developed by Walls & Associates, LLC, and is a private-sector alternative to the Census Bureau's restricted-use Longitudinal Business Database that attempts to track all business establishments operating in the United States. The NETS data are derived from Dun and Bradstreet's (D&B) collection of Duns Marketing Information files that track more than 71 million establishments annually. D&B has a strong reputation for accuracy. For decades, their DUNS identifier, and related establishment data, have been the primary validation mechanism used by the U.S. General Services Administration when individual establishments apply for financial assistance from the federal government or seek federal procurement contracts.³

We chose NETS as our primary measure for several reasons. First, the dataset provides establishment-level longitudinal employment data, allowing us to identify high-growth businesses based on a governance structure and business activity that are of very high relevance to practitioners and policymakers (Davidsson and Wiklund, 2017). Relatedly, employment-based growth measures have been demonstrated to have high concurrent validity (Shepherd and Wiklund, 2009). NETS also contains each establishment's North American Industrial Classification System (NAICS) code, thereby allowing us to analyze the industrial heterogeneity of high-growth establishments.⁴

Using NETS, we operationalize regional HGE as the MSA-level ratio of the number of high-growth establishments per 100,000 establishments. Since employment is the primary metric in the NETS database (sales data are thin and less reliable), an establishment is

³ The federal government codified the use of the DUNS identifier in the Federal Acquisition Regulations in 1998 to identify establishments operating in the U.S. In April 2022, the DUNS number was replaced by a Unique Entity ID (UEI), generated by sam.gov.

⁴ Industry detail is not available for either the *Inc. 5000* or REAI measures of high-growth entrepreneurship.

Table 1
Variable descriptions, summary statistics, and sources.

Variable	Treated MSAs	Comparison MSAs	Description	Source
High-growth entrepreneurship (NETS)	31.278 (42.887)	31.277 (53.528)	The number of high-growth establishments from the National Establishment Time Series database per 100,000 business establishments	National Establishment Time Series (NETS)
High-growth entrepreneurship (<i>Inc. 5000</i>)	40.621 (43.611)	36.569 (41.225)	The number of firms on the annual <i>Inc. 5000</i> list of fastest-growing privately held companies in the U.S. per 100,000 firms.	Inc. Magazine; Business Dynamism Statistics
High-growth entrepreneurship (REAI)	0.511 (1.030)	0.849 (3.675)	Regional Entrepreneurship Acceleration Index. It is the ability of a region to convert entrepreneurial potential into realized growth.	Startup Cartography Project
Pro-market institutions (MEFI)	6.544 (0.643)	6.518 (0.745)	Local Pro-Market Institutions. Values range from 0 to 10, with higher values reflecting more pro-market institutions.	Stansel (2019)
Net migration rate	-155.484 (112.832)	-135.788 (114.217)	Net population flows per 10,000 residents	Internal Revenue Service, Statistics of Income
Regional innovation network	1.360 (1.437)	1.669 (4.809)	The internal social proximity metric for patent authors during a given year.	PatentsView Project
Net job creation rate	0.807 (3.782)	0.737 (3.344)	The ratio of the difference between the number of jobs created and the number destroyed from all establishments during a year (including startups) to the existing number of jobs.	Business Dynamism Statistics (BDS)
Per capita personal income	35.743 (5.143)	38.060 (8.671)	Per capita personal income (\$1,000s), inflation adjusted	Bureau of Economic Analysis
Entrepreneurship rate	20.185 (3.615)	18.636 (3.684)	The ratio of the number of nonfarm proprietors to the number of nonfarm wage employees.	Bureau of Economic Analysis
R&D spending per establishment	10.446 (26.597)	14.058 (35.230)	The sum of real R&D expenditures by colleges and universities and federally funded R&D facilities (in \$1000s), normalized by the number of establishments.	National Science Foundation; BDS
VC availability	0.989 (2.292)	1.463 (4.797)	Inflation-adjusted venture capital investments (in \$1000s), normalized by the number of establishments.	Dow Jones Venture Sources; BDS
Establishment entry rate	9.926 (2.010)	9.397 (1.861)	Count of establishment entrants in year t divided by the average count of employment active establishments in year t and year t-1	Business Dynamism Statistics
Job mobility	131.489 (39.630)	120.022 (47.001)	Number of stable job-to-job transitions per 10,000 workers	Longitudinal Employer-Household Dynamics
Population density	171.443 (117.040)	271.851 (336.679)	Population per squared kilometer	Census Bureau

Notes: The treated MSAs include 17 MSAs in Oregon, Montana, South Carolina, and New Hampshire from 2004 to 2014. Descriptive statistics for the comparison MSAs are based on Comparison Group A shown in Fig. 2. This comparison group includes 166 MSAs. Descriptive statistics using Comparison Groups (A + B) or (A + B + C) do not differ in any meaningful way from the figures shown in this table. Standard errors are shown in parentheses below the mean values. Counts of federal regulations are excluded from this table because the level of analysis differs (federal by industry versus MSA). The section titled “Moderator Variables” describes the regulatory burden measures and provides descriptive statistics.

defined as high growth if its employee count grows at least 20 % per year for three consecutive years. As such, our metric captures relative rather than absolute employment growth (Shepherd and Wiklund, 2009). This is an important consideration given that variation in absolute growth is driven by business size effects, whereas variation in relative growth is subject to baseline effects (Delmar, 1997). Following recommendations by OECD, to control for small baseline effects, we require that an establishment must have at least 10 employees at the beginning of the observation period to be considered high-growth (Daunfeldt et al., 2014; Delmar et al., 2003). For each year, we assigned the number of high-growth establishments located in an MSA to the beginning of the three-year growth period used to derive it (e.g., we assigned the 2018 data, which is based on growth over the period 2014–2017, to 2014). We also derive industry specific local HGE measures using the NAICS codes available in the database. Fig. 2 depicts the mean HGE rate by MSA based on the NETS data. Although the final empirical sample includes 270 MSAs (discussed in more detail later), we illustrate mean HGE rates in Fig. 2 for all 382 MSAs.⁵

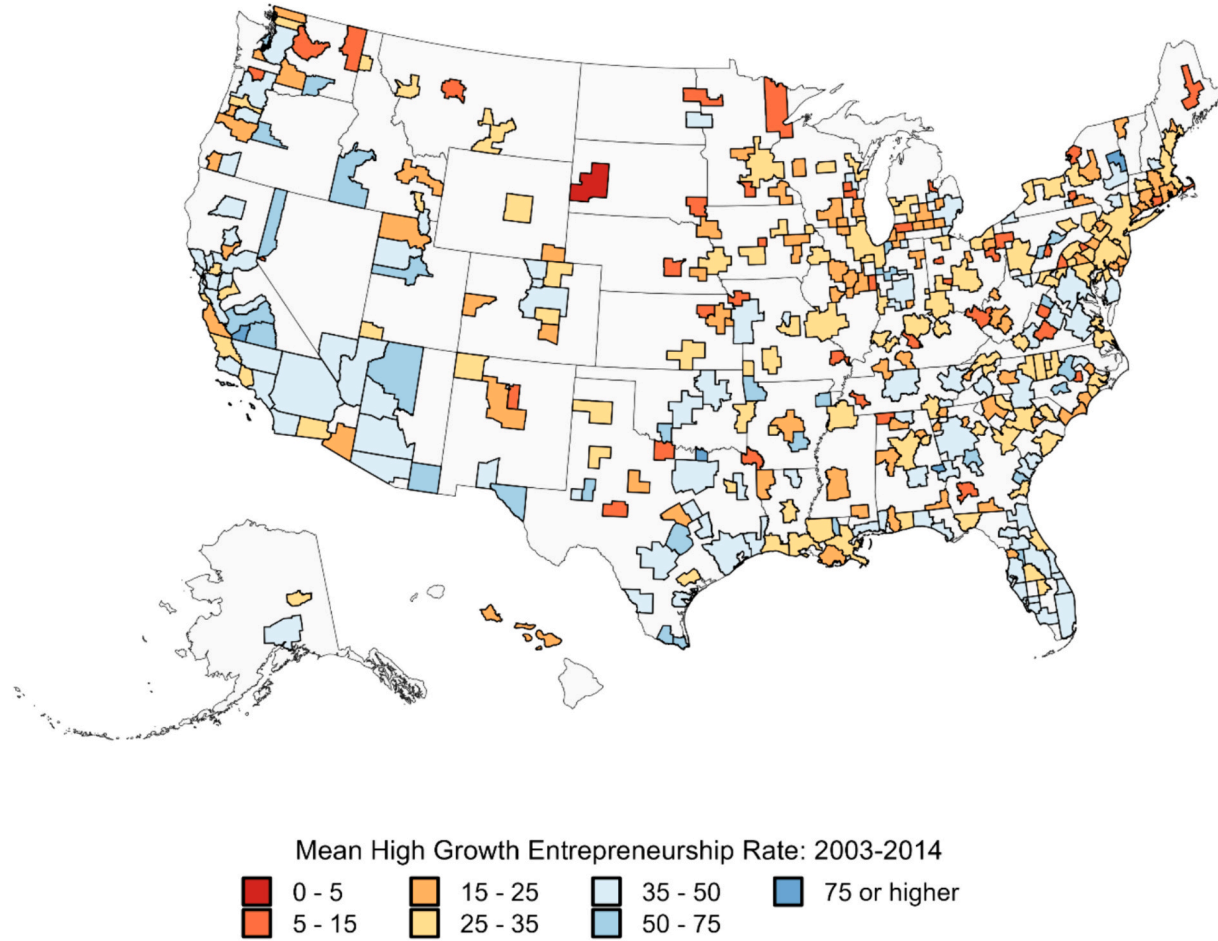
Following Araki et al. (2024), we use as a secondary, revenue-based measure the ratio of firms listed in the Inc. 5000 ranking of the fastest-growing privately held companies in the U.S. per 100,000 firms, as reported by the U.S. Census Bureau's Business Dynamics Statistics (BDS) database. While relative revenue-based measures of growth have been shown to exhibit low convergent validity with alternative measures of venture growth (Delmar et al., 2003; Kiviluoto, 2013; Shepherd and Wiklund, 2009), it is nonetheless a widely used metric with a high degree of generality (Davidsson and Wiklund, 2017). To be eligible for inclusion in the *Inc. 5000*, firms must be U.S. based, independent, and structured as a for-profit company. They must also have experienced annual revenue growth of at least

⁵ MSA definitions are based on the Office of Management and Budget's 2013 delineations.

Table 2
Correlation matrix.

	Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	HGE (NETS)	1.000													
2	HGE (<i>Inc. 5000</i>)	0.063	1.000												
3	HGE (REAI)	0.033	0.163	1.000											
4	R&D spending per establishment	0.011	0.117	0.014	1.000										
5	Net job creation rate	0.064	0.016	0.025	0.002	1.000									
6	Entrepreneurship rate	0.029	0.122	0.006	−0.089	−0.100	1.000								
7	VC availability	0.009	0.276	0.106	0.238	−0.048	0.095	1.000							
8	Net migration rate	0.086	0.115	0.046	−0.159	0.048	0.021	−0.020	1.000						
9	Regional innovation network	0.010	0.228	0.091	0.037	−0.040	−0.019	0.109	0.117	1.000					
10	Per capita personal income	0.012	0.307	0.163	0.027	−0.060	0.319	0.288	0.118	0.211	1.000				
11	Establishment entry rate	0.060	0.155	0.098	−0.028	0.419	0.254	0.053	−0.224	0.001	−0.042	1.000			
12	Population density	−0.004	0.317	0.154	−0.005	−0.041	0.092	0.246	0.116	0.219	0.523	0.071	1.000		
13	Job mobility	0.030	0.152	0.130	−0.046	0.270	−0.078	−0.065	0.074	0.131	0.041	0.368	−0.094	1.000	
14	Pro-market institutions (MEFI)	0.003	0.185	0.128	0.011	0.047	−0.017	0.100	−0.119	−0.018	0.239	0.142	0.060	0.143	1.000

Notes: Counts of federal regulations are excluded from this table because the level of analysis differs (federal versus MSA). The section titled “Institutional Contingency Results” describes the regulatory burden measures and provides descriptive statistics.



Source: Authors' calculations from the National Establishment Time Series database.

Fig. 2. Mean high-growth entrepreneurship rates.

20 % over the previous three years (three-year revenue growth rate of 72.8 %), at least \$100,000 annual revenue at the beginning of the three-year growth period (to minimize baseline effects), and at least \$2 million in revenue by the end of it.⁶ This eligibility criteria aligns with the definition of a “gazelle” (Birch, 1995). We then normalized the number of high-growth firms by the total firm population that year. Consistent with our primary HGE measure, the ratio of high-growth firms located in an MSA are assigned to the beginning of the three-year growth period used to derive it.

As a final robustness check, we also measure local high-growth entrepreneurship using the Regional Entrepreneurship Acceleration Index (REAI) from the Startup Cartography project. This variable, developed by Andrews et al. (2022), is derived from a predictive analytics approach that estimate the number of new business registrations (i.e., establishments) in a region that are expected to later achieve a significant growth outcome, based on a startup's characteristics. A region's REAI score is the fraction of potential significant growth outcomes that are converted into actual outcomes. A region's REAI provides a measure of how well a region's entrepreneurial ecosystem accelerates startups after founding into significant growth outcomes. The REAI provides an alternative perspective on HGE compared to the NETS or *Inc. 5000* measures because it is focused on startups and considers a “successful outcome” if it either achieves an initial public offering (IPO) or is acquired at a positive valuation within 6 years of registration (Andrews et al., 2022). One limitation of the REAI variable is that it is only available at the MSA level until 2012, so we have fewer post-treatment observations to assess the impact of weakening NCA enforceability relative to the NETS and *Inc. 5000* measures.

3.3. Friction-reducing labor market reform

We operationalize friction-reducing regional labor reforms through the lens of the weakening legal enforcement of non-compete agreements (NCAs). In our analysis, we examine the impact of changes in NCA enforceability as identified by Ewens and Marx (2018). Such changes have traditionally been enacted by state-level judiciaries in the U.S. Thus, we use changes in state-level enforceability date between 1985 and 2016, categorized as either weakening, strengthening, or no change in enforceability (Ewens and Marx, 2018). For our study period (2004–2017), and to assess the treatment effect in a Difference-in-Differences (DID) framework, we consider MSA in states that weakened the enforceability of NCAs during our observation period. As depicted in Fig. 3, four states – Oregon, Montana, South Carolina, and New Hampshire – did so. MSAs located in these states are considered our ‘treated units’. This approach allows us to understand how reducing the enforceability of NCAs, and thus reducing labor market frictions, influences the emergence of high-growth ventures. ‘Comparison unit’ MSAs are described in detail in the *Methods* section.

3.4. Moderator variables

Our model includes two moderating institutional variables: federal industry regulatory expansion and local pro-market institutional change. Following recent entrepreneurship studies (Bennett, 2021a, 2021b; Wagner and Bologna Pavlik, 2020), we operationalize pro-market institutional change using data from the metropolitan area economic freedom index (MEFI). The MEFI is composed of three multi-dimensional sub-indices reflecting limited government, tax freedom, and labor market freedom at the MSA-level (Stansel, 2019). Each sub-index is scored on a 0–10 scale, with higher values reflecting stronger pro-market institutions, and the MEFI is the average of the three underlying measures. Because we employ a fixed effects estimator, which time demeans the data, our estimates depict the effect of pro-market institutional changes on changes in HGE (Kimjeon and Davidsson, 2021).

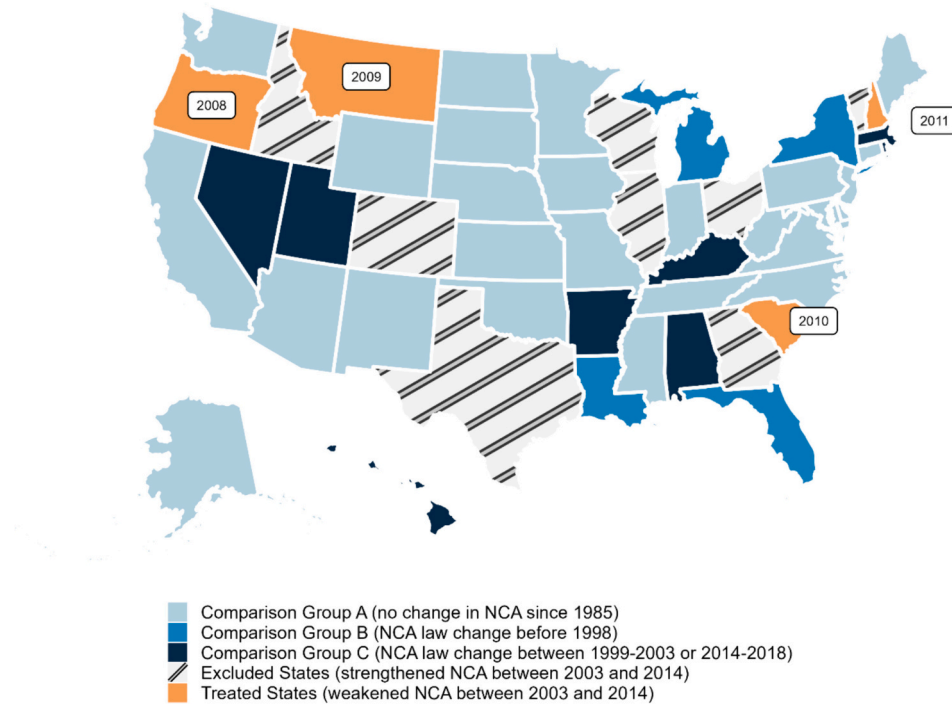
We utilize the Mercatus Center at George Mason University's RegData 4.0, a novel and comprehensive database that provides the first industry-specific quantification of federal regulation in the U.S. This database constructs counts of (federal) industry-level regulations by applying text analysis and machine learning tools to the *Code of Federal Regulations*. At a high level, the RegData provides counts of key words such as shall, must, may not, required, or prohibited to provide a measure of regulatory burden for 15 “super-sector” industries from 1970 to 2019 (Al-Ubaydli and McLaughlin, 2017).

The “supersector” (or two-digit NAICS) industries included in the RegData and in our analysis are: Agriculture Forestry Fishing and Hunting (11), Mining Quarrying and Oil and Gas Extraction (21), Utilities (22), Manufacturing (31), Wholesale Trade (42), Retail Trade (44), Transportation and Warehousing (48), Information (51), Finance and Insurance (52), Professional Scientific and Technical Services (54), Administrative and Support and Waste Management and Remediation Services (56), Educational Services (61), Health Care and Social Assistance (62), Arts Entertainment and Recreation (71), and Public Administration (92).

There regulatory burden varies across industries. While the mean number of requirements per industry during our sample period is 64,218 restrictions, this ranges from an average low of 6150 in wholesale trade to a high of more than 207,000 in manufacturing. The regulatory burden has also expanded in every industry during the sample period, growing at an annual average rate of nearly 3.7 % (or about 44 % cumulative growth). Administrative and Support and Waste Management and Remediation Services experienced the smallest cumulative growth in new requirements at 9.7 %, whereas Utilities experienced the highest cumulative growth (160.0 %).

We take the natural log of the annual regulatory count to provide a continuous measure of regulatory burden. Thus, our time-demeaned fixed effects estimates approximate the growth rate of regulatory compliance burden, a valid measure of regulatory expansion.

⁶ Firms must apply to be included in the *Inc. 5000*, potentially introducing selection bias into the data. However, the significant national publicity associated with the annual ranking encourages firms to participate. Existing research suggests that the characteristics of firms included on the Inc. 5 k ranking are consistent with the general literature on high-growth firms and that the spatial distribution of Inc. 5 k firms, although concentrated in MSAs, reflects the overall geographic distribution of U.S. companies (Hathaway, 2018; Li et al., 2016).



Source: Ewens and Marx (2018).

Fig. 3. Changes in state noncompete laws.

3.5. Control variables

To control for a region's economic dynamism, which can be associated with environment more amenable to high-growth firms (Araki et al., 2024; Li et al., 2016), we included the following variables: (i) net job creation rate, the ratio of the difference between the number of jobs created and the number destroyed from all establishments during a year (including startups) to the existing number of jobs; (ii) the rate of new establishment entries during a year; and (iii) the net migration rate (per 10,000 residents), or the net flow of migrants (migrants less emigrants) to a region during a year, and (iv) the region's entrepreneurship rate. We obtained the MSA-level net job creation and establishment entry rate measures from the Census Bureau's Business Dynamic Statistics (BDS) database, derived the net migration measures from Internal Revenue Service's Statistics of Income (SOI) database, and constructed the entrepreneurship rate from the Bureau of Economic Analysis.⁷

We also controlled for the role of research and development (R&D) investments in facilitating HGE using data from the National Science Foundation's public-use datafiles. It includes all R&D expenditures by colleges and universities (via the Higher Education Research and Development Survey) and federally funded U.S. R&D facilities (via the Survey of R&D Expenditures at Federally Funded R&D Centers). We adjusted R&D investments for inflation and normalized by the number of business establishments in a region. Finally, the regional network of inventors has been demonstrated to be an important determinant of HGE (Araki et al., 2024), so we also controlled for this factor.

Additionally, we include two covariates, population density and job mobility, to adjust for agglomeration effects and within-region worker transitions between employers. Population data are from the Census Bureau. Within-region worker mobility is derived from the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) database. It is the average number of employed workers in an MSA each year who transition to another employer in the region without a period of unemployment, plus workers who change employers with a period of non-employment that does not exceed one quarter.

4. Methods

With a sample of states that have weakened NCA enforceability and a sample of states whose NCA enforceability remained unchanged over the same timeframe, we estimate the causal effect of weakening NCAs on HGE using a difference-in-differences (DID)

⁷ The SOI data are based on year-to-year address changes on federal personal income tax returns, so they more accurately reflect the spatial reorganization of households rather than individuals. We used county-to-county address changes, ignoring migrations within the same region.

estimation strategy. With panel data, this model can be expressed as:

$$HGE_{m,t} = \beta_0 + \beta_1 Post_{m,t} + \nu X_{m,t} + \lambda_t + \mu_m + \epsilon_{m,t} \quad (1)$$

where $HGE_{m,t}$ is our measure of high growth entrepreneurship for MSA m in period t , $X_{m,t}$ is a matrix of MSA-level time-varying control variables, λ_t and μ_m denote fixed year and MSA effects, respectively, and $\epsilon_{m,t}$ is an idiosyncratic error term. The year-specific fixed effects adjust for shocks common to all MSAs for a given year (e.g., the Great Recession following the 2007–2008 financial crisis) while the MSA-specific fixed effects adjust for any unobserved time-invariant factors unique to each MSA (e.g., culture) that could be correlated with HGE.

$Post_{m,t}$ is the DID variable of interest; it is an indicator variable that equals unity for MSAs that have weakened NCAs (the treated units) in the post-treatment period. For example, Oregon weakened NCA enforceability in 2008 so $Post_{m,t}$ would equal one for all MSAs in Oregon starting in 2008 (and all subsequent years) and equal zero every year before 2008 (the pre-treatment period). β_1 , the coefficient of $Post_{m,t}$, is a direct estimate of the average difference in the rate of HGE in the treated MSAs (after weakening NCA enforceability) relative to the average difference in the rate of HGE in the comparison group MSAs whose NCA enforceability did not change.

When there is staggered timing (or variable treatment timing) in a DID framework, Goodman-Bacon (2021) shows that estimation of β_1 will be biased unless the treatment effect is constant over time and across all treated units. This is a very strict assumption. In our setting, it means that weakening NCA enforceability has the same effect on HGE in every MSA over time. If the effect differs across MSAs in any way, either in terms of magnitude or timing, then estimating β_1 using conventional two-way fixed effects will lead a biased estimate (Goodman-Bacon, 2021).

The intuition of this result is surprisingly simple. Assume that the “true effect” of weakening NCA enforceability on HGE is positive. If Eq. (1) is estimated using conventional two-way fixed effects, β_1 will be biased towards zero because MSAs in states that weakened NCA enforceability early in the sample (like Oregon) end up becoming part of the comparison group for states whose MSAs are treated later. Since DID is comparing the pre- and post-treatment average of treated units to the pre- and post-treatment average of the comparison units, some of the “true effect” is differenced away from MSAs treated later in the sample because MSAs treated early are part of their comparison group.

To mitigate this source of bias, we use the stacked regression DID estimator proposed by Cengiz et al. (2019). As Baker et al. (2021) show through simulations, the stacked estimator yields an unbiased estimate of β_1 (the true average treatment effect on the treated) in situations where two-way fixed effects perform poorly. The idea underlying the stacked regression approach is the formation of a comparison group for each treatment group cohort that ensures that treated MSAs do not later contaminate the comparison group. Each treatment group cohort's dataset is then converted to relative time so that treatment dates align across cohorts, eliminating the variable timing issue.

We have seventeen treated MSAs in the four states that weakened NCA enforceability; six each in both Oregon and South Carolina, three in Montana, and one in New Hampshire. Our base comparison group, depicted in Fig. 3 as comparison group A, consists of 166 MSAs from the twenty-six states that never changed their NCA enforceability status between 1985 and 2016. Let g denote the Oregon MSAs whose NCA enforceability status changed in 2008. If the treatment date for any treated cohort g is re-defined as period 0, then cohort g 's dataset uses treated and comparison group observations from period $t - 4$ (or 2004) to period $t + 6$ (2014). Similarly, for the cohort of Montana MSAs whose NCA enforceability status changed in 2009, 2009 is defined as period 0 and the cohort's dataset includes treated and comparison observations from $t - 4$ (2005) to $t + 6$ (2015).

This process is repeated for the cohorts of MSAs weakening NCA enforceability in 2010 and 2011. The final dataset is formed by stacking (row concatenating) every individual cohort dataset. This structure does two things. First, it strictly ensures that no treated MSAs are ever part of a comparison group for another group of treated MSAs. Treated units are always compared to “clean comparison” MSAs that never changed their NCA enforceability. Second, it guarantees that every treated cohort is treated in the same relative time period (period 0).⁸

The final methodological concern we must address is the “parallel trends” assumption. Identification in DID designs is valid only if the comparison group is a suitable counterfactual for what *would have occurred* in the treated MSAs if those MSAs never weakened NCA enforceability. Unfortunately, this assumption is untestable since we never observe what would have happened in the treated MSAs if they did not weaken their NCA enforceability. We follow the standard practice of examining whether the trend in HGEs was similar between the treated and comparison units *before* weakening their NCA enforceability (the pre-treatment period) to provide evidence that the comparison regions may be a valid counterfactual. If HGE was on a similar trajectory in treated and comparison MSA before NCA enforcement was weakened, then it is reasonable to assume the comparison MSAs reflect a valid counterfactual for the treated MSAs.

Fig. 4 plots the (raw) mean HGE rates for treated and comparison units, demeaned by their respective pre-treatment averages, using the NETS HGE measure. Relative period 0 denotes the treatment year for every treated cohort of MSAs. The raw sample mean across treated and comparison MSAs follow the same general trajectory. There are, however, some observable differences in the raw data, so we also formally test for pre-treatment differences across groups in a regression setting. This allows us to leverage covariates that could adjust for observable differences in HGE between treated and comparison MSAs and improve the “fit” between treated and comparison

⁸ An important caveat of the stacked regression DID design is the need to have year and MSA-fixed effects for each cohort (Cengiz et al., 2019). We refer to these as (year $\times$ cohort) and (MSA $\times$ cohort) fixed effects in the regression notes.

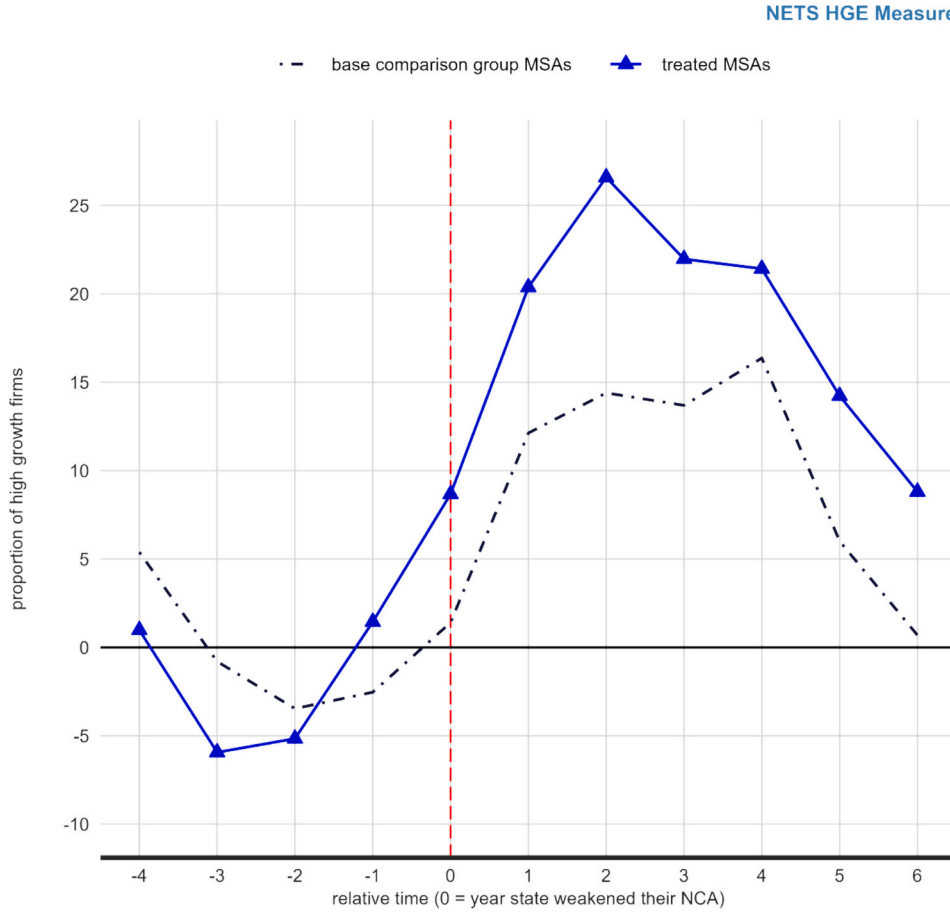


Fig. 4. Trend growth in raw high growth entrepreneurship.

MSAs.

Following the strategies utilized by [Muralidharan and Prakash \(2017\)](#) and [Portillo and Wagner \(2021\)](#), we estimate an alternative version of Eq. (1) to formally test for pre-treatment differences in HGE trend rates between the treated and comparison MSAs. This is more rigorous than simply plotting the raw data. This specification can be expressed as:

$$HGE_{m,t} = \beta_0 + \beta_1 Post_{m,t} + \nu X_{m,t} + \lambda_t + \mu_m + \omega_1 (pre_{m,t} \bullet T) + \omega_2 (pre_{m,t} \bullet T^2) + \epsilon_{m,t} \quad (2)$$

where T denotes a linear trend and $pre_{m,t}$ equals unity for treated MSAs in the pre-treatment period (and equals zero otherwise). The terms $\omega_1 (pre_{m,t} \bullet T)$ and $\omega_2 (pre_{m,t} \bullet T^2)$ allow the treated MSAs HGE rates to follow a different quadratic trend relative to the comparison MSA.

in the pre-treatment period.

Suppose the comparison MSAs are a perfect counterfactual for the treated MSAs. This implies that they would have identical pre-treatment trends in HGE. If this were indeed the case, then ω_1 and ω_2 in Eq. (2) would equal zero and Eq. (2) would reduce to Eq. (1). We estimate Eq. (2) for every specification and conduct an F-test for the joint significance of ω_1 and ω_2 . If we fail to reject the null hypothesis that these parameters jointly equal zero, then this is evidence that there were not meaningful differences in HGE between the treated and comparison MSAs before NCA enforceability was weakened, conditional on the covariates and fixed effects in the regression. If there are no statistically significant differences in HGE between treated and comparison MSAs before NCA enforceability was weakened, then this is the strongest evidence one can offer that the comparison MSAs are a suitable counterfactual for what would have happened in treated MSAs if NCA enforceability was unchanged. We performed all statistical analyses using R.

5. Empirical results

5.1. Baseline results

[Table 3](#) reports our main results. We report a total of nine regressions that vary the high-growth entrepreneurship measure and the

Table 3

Stacked difference-in-differences regression results: main results.

Variable	Model 1A	Model 1B	Model 1C	Model 2A	Model 2B	Model 2C	Model 3A	Model 3B	Model 3C
post (H1)	9.532*** (3.211)	10.185*** (3.078)	8.779*** (2.911)	5.808** (2.837)	6.421** (2.946)	5.622* (3.161)	0.123*** (0.031)	0.153*** (0.030)	0.164*** (0.030)
pro-market institutions (MEFI)	5.861 (5.924)	0.052 (4.890)	−3.024 (4.134)	2.400 (3.084)	−0.730 (2.432)	−4.424 (3.596)	0.107 (0.073)	0.062 (0.054)	0.087* (0.047)
R&D spending per establishment	−0.374*** (0.053)	−0.300*** (0.047)	−0.298*** (0.047)	−0.153*** (0.025)	−0.158*** (0.022)	−0.153*** (0.023)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
VC availability	0.022 (0.083)	0.130 (0.086)	0.194** (0.090)	−0.039 (0.061)	−0.016 (0.064)	0.083 (0.079)	0.001 (0.003)	0.002 (0.003)	0.001 (0.003)
net job creation rate	0.266 (0.405)	0.134 (0.279)	0.148 (0.247)	0.165 (0.108)	0.067 (0.094)	0.172* (0.092)	0.003 (0.004)	0.001 (0.003)	0.001 (0.003)
net migration rate	−0.026 (0.016)	−0.027** (0.014)	−0.009 (0.012)	−0.062*** (0.010)	−0.046*** (0.009)	−0.039*** (0.007)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
entrepreneurship rate	1.860 (1.266)	2.062** (1.020)	1.588 (0.988)	2.961*** (0.592)	2.225*** (0.379)	1.755*** (0.427)	0.079*** (0.019)	0.046*** (0.014)	0.032*** (0.012)
regional innovation network	−0.254** (0.106)	−0.087 (0.095)	−0.125 (0.105)	0.214** (0.086)	0.169** (0.076)	0.161** (0.074)	−0.001 (0.001)	0.000 (0.001)	0.001 (0.001)
per capita personal income	−1.489*** (0.401)	−1.269*** (0.320)	−1.341*** (0.287)	−0.858*** (0.241)	−0.874*** (0.207)	−0.553*** (0.212)	0.005 (0.005)	0.011** (0.004)	0.007* (0.004)
establishment entry rate	1.892* (1.129)	1.366 (0.997)	1.158 (0.847)	1.467*** (0.428)	1.614*** (0.369)	−0.058 (0.543)	0.018 (0.014)	0.024** (0.011)	0.013 (0.010)
population density	0.161*** (0.052)	0.076 (0.052)	0.083* (0.049)	0.242*** (0.056)	0.295*** (0.041)	0.291*** (0.038)	0.002** (0.001)	0.002*** (0.001)	0.002*** (0.001)
job mobility	−0.069 (0.052)	−0.026 (0.054)	0.022 (0.039)	0.122*** (0.038)	0.118*** (0.032)	0.048 (0.034)	−0.001** (0.001)	−0.001 (0.001)	0.001 (0.001)
HGE Measure	NETS	NETS	NETS	<i>Inc. 5000</i>	<i>Inc. 5000</i>	<i>Inc. 5000</i>	REAI	REAI	REAI
N	6954	8930	10,260	6954	8930	10,260	5280	6776	7796
Number of treated MSAs	17	17	17	17	17	17	17	17	17
Number of comparison MSAs	166	218	253	166	218	253	166	218	253
Fig. 3 Comparison Group States	A	A + B	A + B + C	A	A + B	A + B + C	A	A + B	A + B + C
p(differential pre-trends)	0.772	0.656	0.690	0.424	0.356	0.456	0.859	0.898	0.884
p(differential pre-trends)	0.772	0.656	0.690	0.424	0.356	0.456	0.695	0.891	0.906

DV = high-growth entrepreneurship. All models include fixed (MSA × cohort) and (year × cohort) effects, as well as an intercept term. ***0.01; **0.05; *0.10. Standard errors are clustered at the state and MSA dimensions. The HGE REAI measure enters the regression as log(HGE REAI + 1).

comparison group. Models using the NETS HGE measure are reported in columns 1A, 1B, and 1C, models using the *Inc. 5000* HGE measure are shown in columns 2A, 2B, and 2C, and models 3A, 3B, and 3C report estimates using the REAI HGE measures.

As illustrated in Fig. 3, different states have changed their NCA enforceability over time and we leverage this variation to construct several alternative comparison groups as a robustness check. The baseline comparison group, reported as column A in Table 3, uses 166 MSAs located in group A states (26 in all) as shown in Fig. 3. NCA enforceability in these states has remained unchanged since at least 1985. Second, regression results labeled as column B expands the baseline comparison group by including 52 additional MSAs from the four states that changed their NCA enforceability between 1985 and 1998 (more than 5 years before our sample begins). These four states are labeled as group B in Fig. 3. Finally, the largest comparison group, labeled as column C, expands comparison group B by adding 35 MSAs from states that changed their NCA enforceability within 5 years of the start or end of our sample. These eight states are labeled as group C in Fig. 3. All told, comparison group A consists of 166 MSAs, comparison group B includes 218 MSAs, and comparison group C is composed of 253 MSAs.⁹

Standard errors are multi-way clustered at the state and MSA level (Cameron et al., 2011). This allows for unknown correlation within the same MSA over time and across MSAs located in the same state. The row labeled “p(differential pre-trends)” shows the *p*-value from the F-tests for differential pre-treatment HGE trends between treated and comparison MSAs. We find no evidence of significant differences prior to weakening NCA enforceability, providing corroborative evidence for the parallel trends assumption.¹⁰

⁹ Eight states, shown with a stripe pattern in Figure 3, are excluded from the empirical analysis because they strengthened their NCA enforceability at some point during our sample period (2003 to 2014). Multi-state MSAs are assigned to the state where its most populated county is located. For instance, the Cincinnati OH-KY-IN MSA is assigned to Ohio because Hamilton county is in Ohio.

¹⁰ A common alternative approach to probing the potential validity of the parallel trends assumption is to estimate and plot the results of an event study regression. Rather than having a single difference-in-differences coefficient, the event studies estimate a difference-in-differences effect each period to visualize pre- and post-treatment dynamics. If any of the difference-in-differences coefficients are significantly different from zero before treatment, then this would cast doubt on the validity of the parallel trends assumption being true. We also estimated event study versions of each specification in Table 3, omitting the indicator for relative time period −1. These additional plots, available upon request, also show no evidence of any significant differences between the treated and comparison MSAs in the pre-treatment period.

The coefficient for ‘post’ is the DID estimate of interest. The effect is statistically significant at the 5 % level or better in eight of the nine specifications. Using our primary NETS HGE measure, we find that the rate of high-growth establishments is between 8.7 and 10.1 points *higher* in MSAs after enforceability is weakened. The estimated magnitude is slightly smaller for the *Inc. 5000* HGE measures, where we find weakening NCA enforceability increases the high-growth firm rate (per 100,000) between 5.6 and 6.4. Relative to the mean HGE rates in treated MSAs of 31.3 (NETS) and 40.6 (*Inc. 5000*), the point estimates imply that HGE rates increase between 13 and 16 % using *Inc. 5000* and between 27 and 32 % using NETS.

Finally, we also find that weakening NCA enforceability significantly increases a region's ability to convert potential high-growth establishments to actual high-growth businesses, as evidenced by the REAI HGE measures in columns 3A, 3B, and 3C. Because this variable is reported as the ratio of converted to potential high-growth outcomes, we log transform it using $\log(y + 1)$ in the regressions. This is a common transformation that reduces skewness and makes for an easier interpretation because $(100 \times \beta)$ is the approximate percentage change (Bellemare and Wichman, 2020).¹¹ Across models 3A, 3B, and 3C, we find that weakening NCA enforceability increases a region's potential-to-actual high-growth conversion rate between 12.0 and 16.4 %.

Despite employing three growth indicators that exhibit no convergent validity (pairwise correlations all less than 0.1) and reflect different organizational processes (Daunfeldt et al., 2016; Shepherd and Wiklund, 2009), our results provide substantial support for H1, suggesting that reduced enforceability of NCAs facilitates more high-growth entrepreneurship. However, our institutional contingency model points towards important institutional contingencies that influence the scope of HGE enablement resulting from this important labor market reform. We describe our results from these analyses next.

5.2. Institutional contingency results

We next leverage the industry detail in the NETS database to examine whether the sectoral scope of HGE enablement varies in industries subjected to higher or lower levels of regulatory compliance burden. As suggested by H2, we anticipate that the strong positive causal effect that weakening NCA enforceability has on HGE will be negatively moderated in industries facing greater regulatory expansion. We measure regulatory expansion at the industry level (two-digit NAICS) using the RegData 4.0 database, which provides annual measures of regulatory burden for fifteen industries.¹² For each industry for which RegData are reported, we construct HGE measures by MSA and year using the NETS database.¹³

The simplest method to compare the effect of NCA enforceability moderated by regulatory expansion, at the individual industry dimension, is to combine (or “stack”) the individual industry regressions into a single, large regression. Suppose, for instance, one wanted to know if the treatment effect of reducing NCA enforceability was different in industry A from all other industries. This is easily done by, first, stacking the fifteen industry datasets into a single dataset, where only the dependent variable differs across datasets because it measures the *industry's* HGE rate. Next, an indicator variable equaling unity if the dataset is industry A can be interacted with the ‘post’ indicator. If the industry A indicator is statistically different from zero, then the estimated treatment effect of NCA enforceability differs in that industry from the overall average effect across all industries.¹⁴

Estimates of NCA reform-regulatory expansion interaction models are presented in Table 4. Because these regressions measure HGE at an industry level, we introduce two new control variables. First, we include the industry's total factor productivity, from the Bureau of Labor Statistics, to adjust for industry-level productivity changes over time. Second, we include a measure of industry employment concentration in the MSA that is equal to the share of overall industry employment from the region's four largest establishments, calculated from the NETS database. Controlling for industry productivity and local concentration, which serve as proxies for the capital intensity and competitiveness of local industrial structures, enhance our confidence that the effects for our regulation variables reflect the impact of regulation rather than other structural industrial characteristics.

Models 1A, 1B, and 1C provide tests of H2. Consistent with the results in Table 3, columns labeled A, B, and C differ only in the states and MSAs that constitute the comparison groups. Consistent with H2, we find that higher federal regulatory expansion dampens the positive effect of weakening NCA enforceability on HGE, as evidenced by the coefficients on the term “post * log(regulation)”. This effect is significant at the 1 % level in each specification.

In terms of magnitude, the typical industry faces a regulatory burden equal to 64,000 compliance requirements (counts of key words such as shall, must, may not, required, or prohibited), and the standard deviation across industries is 51,200. Across all three specifications (Models 1A–1C), the estimates imply that a one standard deviation in an industry's regulatory expansion (relative to the

¹¹ Since the $\log(0)$ is undefined, $\log(y + 1)$ is used to retain zero valued observations. Bellemare and Wichman (2020) also propose an inverse hyperbolic sine transformation as an alternative to $\log(y)$ when zero values are present. We also estimated models 3A, 3B, and 3C using the inverse hyperbolic sine of HGE REAI and find very similar results. In those specifications, we find that potential-to-actual high-growth conversion rates increase between 16 and 20 % after weakening NCA enforceability.

¹² The two-digit NAICS codes for the industries included in RegData 4.0 are: 11, 21, 22, 31, 42, 44, 48, 51, 52, 54, 56, 61, 62, 71, and 92.

¹³ We are unable to perform analogous estimates using the *Inc. 5000* or REAI HGE measures because these data do not report NAICS codes.

¹⁴ In our application, each industry's dataset is itself a “stacked difference-in-differences” dataset following our main results in Table 3.

Table 4
Stacked difference-in-differences regression results: industry results.

Variable	Model 1A	Model 1B	Model 1C	Model 2A	Model 2B	Model 2C
post	96.341 (68.138)	119.186* (67.030)	153.975** (65.676)	1245.694*** (372.498)	1269.148*** (375.146)	1298.826*** (379.402)
pro-market institutions (MEFI)	−33.805** (15.987)	−34.710*** (13.195)	−20.979** (10.048)	−33.942** (15.985)	−34.813*** (13.193)	−21.053** (10.048)
log(regulation)	−4.087*** (1.104)	−3.143*** (0.894)	−2.603*** (0.792)	−4.096*** (1.103)	−3.150*** (0.894)	−2.609*** (0.792)
post * MEFI	−0.990 (6.524)	−3.022 (6.474)	−7.134 (6.547)	−178.414*** (58.546)	−180.539*** (58.989)	−183.862*** (59.692)
post * log(regulation) (H2)	−7.931*** (2.986)	−8.854*** (2.876)	−9.284*** (2.823)	−115.496*** (32.246)	−116.473*** (32.376)	−116.419*** (32.410)
post * MEFI * log(regulation) (H3)				16.606*** (5.086)	16.614*** (5.108)	16.540*** (5.110)
industry total factor productivity	2.434*** (0.658)	1.891*** (0.531)	1.630*** (0.466)	2.431*** (0.658)	1.888*** (0.531)	1.627*** (0.466)
industry employment concentration	0.107 (0.103)	0.086 (0.081)	0.074 (0.071)	0.106 (0.103)	0.085 (0.081)	0.073 (0.071)
R&D spending per establishment	−0.152** (0.067)	−0.137** (0.050)	−0.134** (0.048)	−0.152** (0.067)	−0.137*** (0.050)	−0.134*** (0.048)
VC availability	0.054 (0.178)	0.033 (0.165)	0.033 (0.157)	0.053 (0.178)	0.033 (0.165)	0.032 (0.157)
net migration rate	−0.096*** (0.023)	−0.084*** (0.020)	−0.049*** (0.017)	−0.096*** (0.023)	−0.085*** (0.020)	−0.049*** (0.017)
regional innovation network	1.977*** (0.555)	1.867*** (0.510)	1.793*** (0.484)	1.977*** (0.556)	1.867*** (0.510)	1.793*** (0.484)
per capita personal income	−1.153 (0.996)	−0.888 (0.735)	−0.595 (0.635)	−1.151 (0.996)	−0.887 (0.735)	−0.594 (0.635)
establishment entry rate	−0.543 (2.546)	−0.061 (2.085)	0.064 (1.732)	−0.545 (2.546)	−0.061 (2.085)	0.065 (1.732)
population density	−0.124 (0.204)	−0.108 (0.141)	−0.061 (0.113)	−0.125 (0.204)	−0.108 (0.141)	−0.061 (0.113)
net job creation rate	3.521*** (1.061)	2.340*** (0.791)	2.024*** (0.700)	3.526*** (1.060)	2.343*** (0.791)	2.027*** (0.700)
job mobility	−0.399*** (0.107)	−0.337*** (0.093)	−0.248*** (0.071)	−0.398*** (0.106)	−0.337*** (0.093)	−0.248*** (0.071)
HGE Measure	NETS	NETS	NETS	NETS	NETS	NETS
N	76,473	98,209	112,817	76,473	98,209	112,817
Number of treated MSAs	17	17	17	17	17	17
Number of comparison MSAs	166	218	253	166	218	253
Fig. 3 Comparison Group States	A	A + B	A + B + C	A	A + B	A + B + C

DV = high-growth entrepreneurship. All models include fixed (MSA × cohort) and (year × cohort) effects, as well as an intercept term. ***0.01; **0.05; *0.10. Standard errors are clustered at the state and MSA dimensions.

mean) reduces the high-growth establishment rate from weakening NCA enforceability between 2.0 and 2.3.¹⁵

The final set of estimates in Table 4, Models 2A, 2B, and 2C, provide tests of H3, which suggests that local pro-market institutional changes compensate the negative moderating effect of federal regulatory expansion on the positive relationship between reduced NCA enforceability and HGE. The coefficient of interest for these tests are reported as “post * log(regulation) * MEFI”. This captures the moderating effect of pro-market institutional changes, operationalized by the metropolitan area economic freedom index (MEFI), on federal industry regulatory burden.

The positive and statistically significant coefficient estimates across all specifications lends support that local pro-market institutional changes counteract the disabling effects of federal regulatory. Within the treated MSAs, the mean and standard deviation of MEFI is 6.54 and 0.64, respectively. Relative to an average regulatory burden, our estimates indicate that a one standard deviation increase in the regulatory expansion reduces the high-growth establishment rate from weakening NCA enforceability between 2.1 and 2.4, all else equal. If a local region increases its pro-market institutions by one standard deviation (0.654), then the high-growth establishment rate would be expected to increase by 2.8 rather than decline due to the higher regulatory burden.

¹⁵ The change in regulatory expansion, relative to an average industry, from a one standard deviation increase in requirements is given by: $0.2552 = \log(64,000 + 51,200) - \log(64,000)$. From Model 1A, the coefficient of “post * log(regulation)” is -7.931 . Thus, a one standard deviation increase in regulatory burden (relative to the mean industry) would reduce the high-growth establishment rate from weakening NCA enforceability by 2.0 ($=0.2552 * -7.931$). In addition, the point estimates from Model 1A imply a regulatory inflection point of 180,000, which is roughly 2.3 standard deviations above the mean regulatory burden. Regulatory burdens at or above this threshold are high enough to reverse the effect of weakening NCAs.

6. Discussion

Our central insight is that a significant institutional change that reduces frictions in the labor market may serve as an external enabler of regional high-growth entrepreneurship (HGE). Specifically, we examine how legal reforms that reduce the enforceability of non-compete agreements (NCAs) may enable HGE through transaction costs mechanisms, including compression, conservation, and human resource expansion. Our baseline causal estimates are in line with our first hypothesis (H1) suggesting an enabling effect on HGE. The range of our baseline estimates indicates an effect that translates to the creation of 8.7 to 10.1 new high-growth establishments per 100,000 incumbent establishments in the analyzed period.

However, these results do not hold in all conditions. The enabling effect of the analyzed labor market reform on HGE is dependent on the set of broader institutional conditions facing economic actors, ultimately limiting the sectoral and geographic scope of our focal external enabler. Our multi-level institutional contingency model reveals that, consistent with H2, the positive effect of NCA reform on HGE is weakened in sectors facing greater federal regulatory expansion, suggesting that the sectoral scope of enablement is limited to industries facing lower regulatory barriers to growth (Eberhart et al., 2017; Eesley, 2016). It further reveals that, in alignment with H3, local pro-market institutional changes may offset the dampening effect of regulatory expansion (Lucas and Boudreaux, 2020). Next, we describe the theoretical contributions of our study.

6.1. Theoretical contributions

6.1.1. Contributions to institutions and entrepreneurship literature

We make two important contributions to the institutions and entrepreneurship literature. First, we advance research on institutional contingencies by demonstrating how institutional changes at various governance levels interact to shape entrepreneurial outcomes, specifically HGE. Most existing studies focus on the joint role of formal and informal institutions at the national level in shaping entrepreneurial outcomes like venture capital activity (Li and Zahra, 2012), innovation (Bennett and Nikolaev, 2021; Ilyas et al., 2023) and new venture creation (Boudreaux et al., 2023; Eesley et al., 2018). Building on Lucas and Boudreaux's (2020) insight that sub-national pro-market institutions may compensate for the negative effect of federal regulation, our findings reveal that while state-level labor market reforms, like reducing NCA enforceability, can enhance HGE, this effect is weakened in industries facing greater federal regulatory expansion. However, this weakening effect is offset by local pro-market institutional changes, demonstrating how institutional forces collectively shape the geography of high-growth entrepreneurship. Our findings not only provide further empirical substantiation of the nascent MPF approach in entrepreneurship (Lucas and Boudreaux, 2020), but also extend it by showing that interdependence of federal regulatory and subnational economic institutional reforms on entrepreneurial outcomes may also be conditional on judiciary-driven institutional reforms. Thus, our multi-level institutional analysis paves the way for future entrepreneurship research to incorporate insights from the MPF approach to examine institutional contingencies across governance levels and provide a richer understanding of the scope and boundary conditions of institutional reforms. This approach could also be adopted to study the entrepreneurial effects of other types of institutional reforms such as occupational licensing or intellectual property reforms (Burton et al., 2019; Campbell et al., 2017).

Second, our study acknowledges that entrepreneurship is a multi-stage process influenced differently by institutional changes across lifecycle stages, including the growth stage. We contribute to an emerging body of research examining the effect of exogenous institutional changes on entrepreneurial venture growth (Eberhart et al., 2017; Eesley, 2016; Venâncio et al., 2020) by demonstrating that regional labor market reforms that reduce frictions (e.g., diminished NCA enforceability) not only facilitate entrepreneurial entry, they also induce regional HGE.

6.1.2. Contributions to the NCA-entrepreneurship literature

While previous studies largely examine the impact of reduced NCA enforceability on new venture formation and survival (Buenstorf et al., 2016; Gilson, 1999; Jeffers, 2024; Marx, 2021; Rauch, 2016; Samila and Sorenson, 2011; Starr et al., 2018), we move beyond this initial focus to examine how these labor market reforms impact high-growth entrepreneurship, a common indicator of entrepreneurship quality (Araki et al., 2024; Sleuwaegen and Ramboer, 2020). As such, our study adds to the cumulative body of NCA-entrepreneurship research, suggesting that weakening NCA enforceability not only enhances the quantity but also the quality of entrepreneurial ventures. More importantly, we contribute the novel insight that the effects of NCA reforms on HGE depend on the broader institutional context. Our study reveals that the positive effects of reducing NCA enforceability on HGE are significantly moderated by other institutional changes, particularly federal regulatory expansion and local pro-market institutional changes, suggesting that future research incorporate insights from MPF theory and carefully consider the institutional landscape across governance levels when examining the entrepreneurial effects of NCA (or other exogenous institutional or policy) reforms. Doing so will provide a more nuanced understanding of the scope of these effects.

6.1.3. Contributions to external enablement framework

The concept of external enablement and the EE framework are gaining traction in entrepreneurship research to explain how diverse external changes create opportunities for new venture creation. In addition to providing further empirical substantiation of the EE framework (Chen et al., 2020; Schade and Schuhmacher, 2022), we build on and extend the EE framework in several important ways. First, we are the first to (the best of our knowledge) apply the EE framework to high-growth entrepreneurship. While much of the EE literature focuses on the triggering role of environmental changes in enabling new venture creation and recent studies examine how environmental changes influence entrepreneurial phenomena such as startup human resource recruitment (Lucas et al., 2023),

entrepreneurial ecosystem emergence (Juma et al., 2023), and corporate innovation (Cestino Castilla et al., 2023), we are among the first to embrace the outcome-enhancing and process shaping roles of an EE, laying the groundwork for future studies to further investigate external enablement in post-entry contexts like venture scaling and performance and investigate the relative important of the various roles of an EE (Kimjeon and Davidsson, 2021).

Our study also calls attention to the importance of considering interdependencies between environmental changes across levels. While most EE studies focus on the role of a single environmental change (Kimjeon and Davidsson, 2021), we demonstrate how multi-level environmental changes interact to shape the scope of entrepreneurial enablement. Our findings reveal that while our focal EE, reduced NCA enforceability, enhances HGE in general, this enablement is much more nuanced and depends on the broader and multi-level set of institutional changes, providing us with a richer understanding of the compounding effects of concurrent EEs across spatial and sectoral boundaries. Our study therefore takes an important step to address recent calls for EE scholars to develop “an improved understanding of the...additive, interactive, and causally interrelated environmental changes” (Davidsson et al., 2023, p. 328; Kimjeon and Davidsson, 2021), providing a pathway to better leverage the EE framework to analyze the interdependence of dynamic environmental forces in shaping entrepreneurial outcomes.

Relatedly, our study highlights how concurrent environmental changes may not only enable but also disable entrepreneurship. We introduce to disabling mechanisms, resource escalation and elongation, to explain how federal regulatory expansion increases transactions costs and diverts scarce resources that could otherwise be directed towards growth, dampening the positive effects of NCA reform on HGE. This not only suggests that interrelated environmental changes may have both additive and subtractive effects on entrepreneurship, it also lays a foundation for the future development of an external disablement framework that can be applied, in conjunction with the EE framework, to provide a richer analytical toolset for examining the broader social implications of environmental changes that have both beneficial as well as harmful effects on entrepreneurial activities (Bennett, 2019, 2021a).

6.2. Practical implications

6.2.1. Policy implications

While writing this article, the Federal Trade Commission (FTC) passed a sweeping rule that would ban the enforceability of NCAs throughout the U.S. for all covered employees, excluding senior executives subject to a current NCA. The controversial rule would also prohibit employers from creating or attempting to enforce new NCAs (Federal Trade Commission, 2024). However, there is considerable uncertainty regarding whether the FTC's ban will be implemented, as it will need to survive legal challenges that started immediately following its announcement by groups opposed to the rule (Crowell and Moring, 2024). Given the large stakes, these legal challenges could continue for years and eventually be overturned by the courts. Our findings have important implications for the ongoing and controversial public debate concerning NCAs. While firms employing them maintain that NCAs provide a valuable source of protection for their intangible knowledge assets and incentivize them to make strategic investments in human capital (Conti, 2014; Garmaise, 2011; Starr, 2019), critics point out that NCAs raise the cost of firm departure and, by doing so, restrict labor mobility (Gilson, 1999; Marx et al., 2009). As noted above, studies have documented that NCAs have adversely affected entrepreneurship by reducing new venture creation (Marx, 2021; Samila and Sorenson, 2011; Starr et al., 2018) and restricting the ability of venture capital-backed startups to professionalize their leadership (Ewens and Marx, 2018). Thus, NCA enforceability involves a trade-off between the interests of incumbent ventures and those of employees and potential entrepreneurs (Jeffers, 2024).

We contribute new causal evidence to this debate by showing that states which have weakened the enforceability of NCAs have experienced an increase in high-growth entrepreneurship, an increasingly important public policy objective. In its announcement of the NCA ban, the FTC touted the potential creation of 8500 new businesses each year from the rule change (Federal Trade Commission, 2024). Based on establishment counts in all 382 MSAs, our baseline estimates imply that eliminating NCAs nationwide may result in the emergence of between 340 and 534 HGEs.¹⁶

However, this estimate requires several important caveats. First, our empirical analysis reflects a reduction in legal enforceability while the FTC rule would prohibit their enforceability altogether. Next, extrapolating our estimates from MSAs to the entire country assumes that the treated MSAs in our analysis are representative of those across the U.S. Additionally, our econometric estimates are based on state-level NCA reforms and may capture, in part, the relocation of individuals or business from states with stronger to those with weaker NCA enforceability. Relocations of this nature should cease to occur if NCAs were prohibited nationwide. Furthermore, as we described above, these effects are contingent on the broader, multi-level institutional environment, suggesting that there is considerable complexity and nuance regarding how NCA reform influences entrepreneurship. Thus, policymakers need to take these interdependencies into account when contemplating labor market reforms that alter the legal enforceability of these controversial contracts.

6.2.2. Practitioner implications

Our findings suggest that entrepreneurs with high-growth aspirations should consider the institutional conditions in prospective locations for their ventures. For prospective entrepreneurs currently subject to a NCA who live in a state that enforces such agreements, the transaction costs associated with launching and growing a new venture are substantial. Individuals in such situations may consider

¹⁶ Using the NETS HGE definition and establishment counts, the typical MSA would be expected to gain 0.9 new high-growth establishments. If one assumes that NCA enforceability applies to all establishments, rather than those with 10 or more employees, then a typical MSA would expect to gain 1.4 new high-growth establishments based on our estimates.

relocating to a state with weaker enforcement of such contracts to lower their transaction costs and expand their access to skilled human capital. They may also want to consider the extent of federal regulation governing their focal industry as well as the local institutional environment. Entrepreneurs that have already launched their ventures and have their eyes set on growth may be hindered in this aspiration by the high costs of attracting high human capital employees imposed by NCAs. Such entrepreneurs may also want to consider relocating to states with weak enforcement of such contracts where their access to human capital resources will be expanded. Importantly, they also should consider the local institutional environment in prospective locations, particularly if they are competing in an industry facing considerable regulatory expansion. Specifically, localities implementing more pro-market institutions in states that do not force NCAs may free up entrepreneurs' time and resources to focus on growing their ventures.

However, if the FTC's ban on NCAs withstands immense legal challenges and lawsuits and eventually goes into effect, then entrepreneurs will no longer face this specific institutional constraint that creates frictions in the labor market that hamper entrepreneurial growth. Nonetheless, the practical implications of our study may generalize to other types of labor market institutions (e.g., non-disclosure agreements, intellectual property enforcement, occupational licensing regulations, hiring and firing regulations) that, by impeding labor mobility, create transaction costs that potentially hinder entrepreneurial venture formation and growth (Burton et al., 2019; Campbell et al., 2017; Feldmann, 2009). Thus, entrepreneurs oriented towards growth need to consider a multitude of institutionalized labor market frictions as part of their venture location decision-making process.

6.3. Limitations & opportunities

Like all studies, ours has several limitations that we believe provide opportunities for future research. First, we follow Ewens and Marx (2018) in coding changes in NCA enforceability as binary and plausibly exogenous events that have a broad scope across sectors. However, others have suggested that NCA institutional regimes vary across multiple dimensions and can be assessed in a more granular fashion. Bishara (2011), for instance, developed a state-level NCA enforceability strength index using several questions from a national survey for the years 1991 and 2009 that has been used in several other studies (Starr, 2019; Starr et al., 2018). It is conceivable, therefore, that changes in NCA enforcement may occur across multiple dimensions and/or may be more fine-grained. Such changes may differentially impact HGE as more and better data become available on the enforceability of NCAs, future research can examine these possibilities.

Next, while we control for a variety of potential confounding variables in our regressions, it is conceivable that other time-varying factors are omitted that may bias our results. We are confident that omitted variable bias is not a major issue because our estimated effect of reduced NCA enforceability on HGE is relatively stable when we introduce control variables and robust to several alternative control groups. Additionally, our results are robust when using alternative measures of HGE that capture different dimensions of growth (Daunfeldt et al., 2014; Delmar et al., 2003). Unfortunately, our alternative measures of HGE are not available by industry, so we are unable to examine the robustness of our institutional contingency findings using these measures. Relatedly, our measures of pro-market institutional changes and regulatory expansion may not fully capture the nuanced differences and complexities of these multi-dimensional constructs. As more and better data become available, it would be worthwhile to examine the robustness of our findings using alternative measures. Lastly, our study focuses on weakening NCA enforceability as an external enabler of HGE and, as such, we did not consider the impact of states that strengthened NCA enforceability. In preliminary analysis, we found evidence of a symmetrical effect, namely that an increase in the enforcement of NCAs is associated with a decrease in HGE. These results are omitted but available upon request.

Author statements

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CRedit authorship contribution statement

Daniel L. Bennett: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Data curation, Conceptualization. **Gary Wagner:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Michael Araki:** Writing – review & editing, Writing – original draft.

Data availability

The authors do not have permission to share data.

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